1 Database management systems

Week 1 · Week summary

Core ideas, lesson by lesson

1 From files to databases

- File-based systems duplicated data, so copies became inconsistent.
- Programs depended on file layouts: a layout change meant rewriting code.
- A DBMS stores data once, describes it, protects it, and shares it.

2 Data, information, and what a database is

- Data + context = information; people turn information into knowledge.
- A database models a mini-world, is coherent, and has a purpose.
- The catalogue stores metadata: the structure and rules of the data.

3 What a DBMS does

- A DBMS defines, constructs, manipulates, and shares databases.
- Database system = DBMS + data + application programs.
- Key advantages: less redundancy, access control, fast queries, recovery, integrity.

4 Who works with a database

- DBA: access, performance, security. Designer: structure, rules, views.
- Analysts specify; application programmers build and maintain.
- End users can be casual, parametric, sophisticated, or stand-alone.

5 The relational model: tables, rows, and columns

- Relation = table, attribute = column, tuple = row.
- Schema = structure; data (instance) = content at a moment.
- Other models exist: hierarchical, network, object-oriented, NoSQL.

6 Keys and relationships

- PK: unique and not null; it identifies each row.
- FK: stores another table's PK; it creates the relationship.
- N:M relationships need a junction table with two FKs.

7 Integrity, transactions, and safety

- Constraints: PRIMARY KEY, FOREIGN KEY, NOT NULL, CHECK.
- Transaction = all or nothing: COMMIT or ROLLBACK.
- Concurrency control, logs, backups, and least privilege keep data safe.

8 Three levels, views, and indexes

- External = user views; conceptual = logical model; internal = physical storage.
- Data independence lets one level change without breaking the others.
- Indexes speed up searches but cost space and update time.

9 First SQL queries

- SELECT columns FROM table WHERE condition.
- ORDER BY sorts; JOIN … ON combines tables through keys.
- UPDATE and DELETE always need a WHERE clause.

10 Design in three stages: the ER model

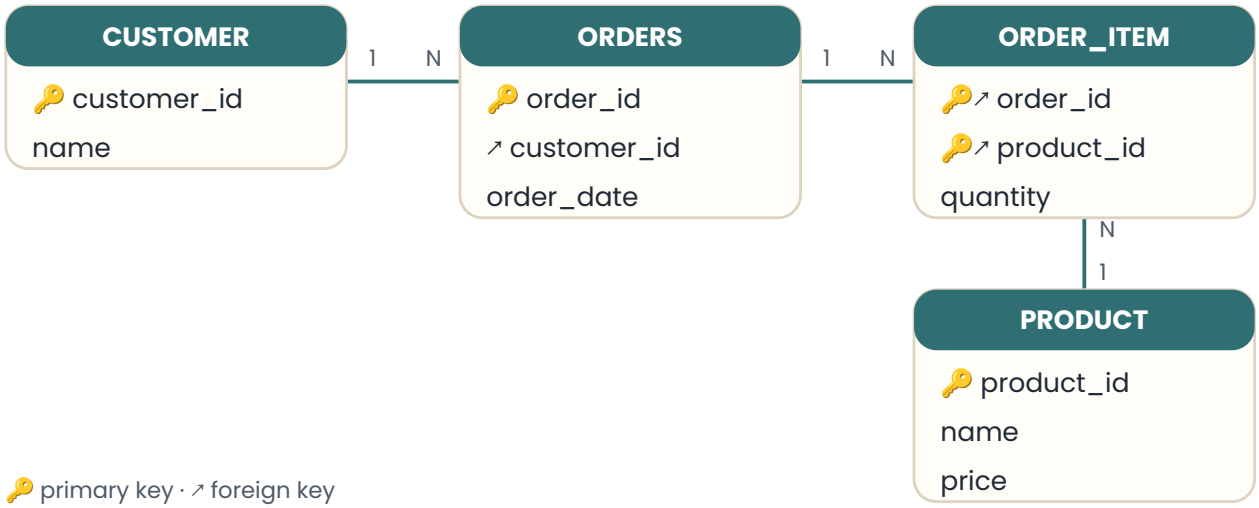
- Conceptual (ER) → logical (tables) → physical (SQL).
- Entities are nouns (rectangles); relationships are verbs (diamonds).
- Attributes: simple, composite, multivalued (double), derived (dashed), key (underlined).
- A weak entity depends on its owner: key = owner key + discriminator.

11 From the ER diagram to tables

- Entity → table; composite → several columns; multivalued → new table; derived → not stored.
- 1:N → FK on the N side; N:M → junction table; 1:1 → FK on either side.
- Keys can be natural, surrogate, or composite.

12 SQL to build and change data: DDL and DML

- DDL: CREATE, ALTER, DROP, TRUNCATE. DML: INSERT, SELECT, UPDATE, DELETE.
- Use DECIMAL for money, VARCHAR for text, DATE for dates.
- ON DELETE: CASCADE, RESTRICT, or SET NULL, chosen per relationship.



Watch out

Mistake	Fix
Confusing the database with the DBMS.	The DBMS is the software; the database is the data it manages.
Modelling many-to-many with a single foreign key.	Add a junction table with both foreign keys.
Thinking a spreadsheet shared by email is a database.	Ask: is there one copy, a described structure, and controlled access?
Confusing the database (the data) with the DBMS (the software).	PostgreSQL is a DBMS; "the store database" is the data it manages.
Storing several values in one cell, such as "red, blue, green".	One value per cell; use another table for lists.
Using a name as the primary key.	Names repeat and change. Use a stable identifier such as customer_id.
Checking rules only in the app's screens.	Also declare them in the database, so every program must follow them.
Running UPDATE without WHERE.	Write the WHERE first, and test it with a SELECT using the same condition.